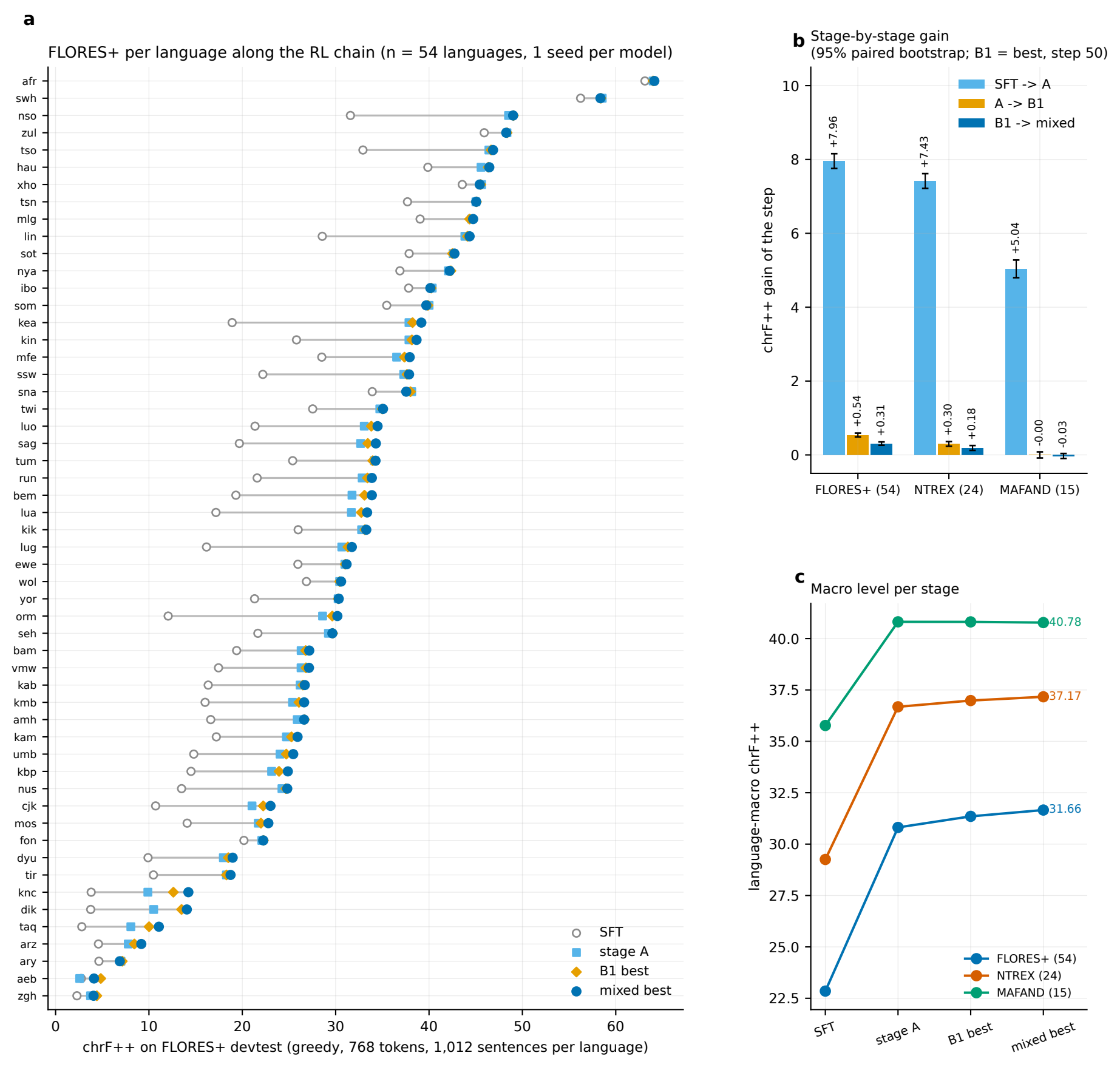
a
FLORES+ per language along the RL chain (n = 54 languages, 1 seed per model)
afr
swh
nso
zul
tso
hau
xho
tsn
mlg
lin
sot
nya
ibo
som
kea
kin
mfe
ssw
sna
twi
luo
sag
tum
run
bem
lua
kik
lug
ewe
wol
yor
orm
seh
bam
vmw
kab
kmb
amh
kam
umb
kbp
nus
cjk
mos
fon
dyu
tir
knc
dik
taq
arz
ary
aeb
zgh
SFT
stage A
B1 best
mixed best
0
10
20
30
40
50
60
chrF++ on FLORES+ devtest (greedy, 768 tokens, 1,012 sentences per language)
b
Stage-by-stage gain
(95% paired bootstrap; B1 = best, step 50)
SFT -> A
A -> B1
B1 -> mixed
chrF++ gain of the step
+7.96
+0.54
+0.31
+7.43
+0.30
+0.18
+5.04
-0.00
-0.03
FLORES+ (54)
NTREX (24)
MAFAND (15)
c
Macro level per stage
language-macro chrF++
40.78
37.17
31.66
FLORES+ (54)
NTREX (24)
MAFAND (15)
SFT
stage A
B1 best
mixed best